

Model Card

13 April 2026

Name	2026 BBO Project by Dave Amin
Type	This project covers 8 functions within a Black Box Optimisation project.
Version	The BBO project started on 29 January 2026, with one submission per week for 12 weeks. Each week therefore creates a new version of the models. As at 13 April 2026, this covers version 10 of the models.
Developer	Dave Amin
Overview	The project is part of the following course at Imperial College: Professional Certificate in Machine Learning and Artificial Intelligence Each function has a description (see below) that describes the objective. Licensing: The project descriptions and initial data were created by Imperial College.
Task / Objective	The task is to maximise the results for all 8 functions. i.e. find the global maximum The students need to avoid being trapped by local maxima.
Strategy	Various models and acquisition functions were used, and these changed as the project progressed week by week. Google Gemini was used as an AI Modelling Assistant.
Results	See below for results, which show how the models have performed.
Intended uses and limitations	The models were designed and used for the above course only. They are purely for educational and review purposes.
Training data	The initial dataset was provided by Imperial College. It is available here: https://github.com/BriarCloudview/Capstone1/tree/initial-data
Evaluation metrics	The purpose of the project is to teach AI principles. Performance results in a real world scenario is not applicable. Evaluation by the course management is expected at the end of the course.
Assumptions, constraints and potential failure modes	It is assumed there are local minima in the search space, and noise, and well as data plateaus which will cause difficulty. The function descriptions assist in identifying these constraints. There is a specific data constraint in that all input values are between 0 and 1. There were various potential failure modes, and the models had to adapt or be changed due to this. Specific problems encountered were: - non-stationarity. It was discovered there were areas where the data was extremely jagged, and had cliffs. - Local Traps. Local maxima were common, and exploration was required to hunt for the global maxima. - Noise. Models had to have their hypermeters tuned due to the noise in the functions.
Ethical considerations	As this is a fictional dataset for educational purposes, no ethical considerations have arisen.

Transparency	The project attempts to be transparent, as it will publish: - The data points submitted - The results obtained - The Final Kernel For example: Final Kernel: $4.26 \times 10^{-2} \times \text{RationalQuadratic}(\alpha=0.0983, \text{length_scale}=2.42) + \text{WhiteKernel}(\text{noise_level}=0.000707)$ Kappa: 3.0 (High Exploration Mode) All of the above are contained within the Python code, which will be available on GitHub. This should allow others to reproduce the same results.
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Function Descriptions

Function	Description
1	Detect likely contamination sources in a two-dimensional area, such as a radiation field, where only proximity yields a non-zero reading. The system uses Bayesian optimisation to tune detection parameters and reliably identify both strong and weak sources.
2	Imagine a black box, or a mystery ML model, that takes two numbers as input and returns a log-likelihood score. Your goal is to maximise that score, but each output is noisy, and depending on where you start, you might get stuck in a local optimum. To tackle this, you use Bayesian optimisation, which selects the next inputs based on what it has learned so far. It balances exploration with exploitation, making it well suited to noisy outputs and complex functions with many local peaks.
3	You're working on a drug discovery project, testing combinations of three compounds to create a new medicine. Each experiment is stored in <code>initial_inputs.npy</code> as a 3D array, where each row lists the amounts of the three compounds used. After each experiment, you record the number of adverse reactions, stored in <code>initial_outputs.npy</code> as a 1D array. Your goal is to minimise side effects; in this competition, it is framed as maximisation by optimising a transformed output (e.g. the negative of side effects).
4	Address the challenge of optimally placing products across warehouses for a business with high online sales, where accurate calculations are costly and only feasible biweekly. To speed up decision-making, an ML model approximates these results within hours. The model has four hyperparameters to tune, and its output reflects the difference from the expensive baseline. Because the system is dynamic and full of local optima, it requires careful tuning and robust validation to find reliable, near-optimal solutions.
5	You're tasked with optimising a four-variable black-box function that represents the yield of a chemical process in a factory. The function is typically unimodal, with a single peak where yield is maximised. Your goal is to find the optimal combination of chemical inputs that delivers the highest possible yield, using systematic exploration and optimisation methods.
6	You're optimising a cake recipe using a black-box function with five ingredient inputs, for example flour, sugar, eggs, butter and milk. Each recipe is evaluated with a combined score based on flavour, consistency, calories, waste and cost, where each factor contributes negative points as judged by an expert taster. This means the total score is negative by design. To frame this as a maximisation problem, your goal is to bring that score as close to zero as possible or, equivalently, to maximise the negative of the total sum.

7	You're tasked with optimising an ML model by tuning six hyperparameters, for example learning rate, regularisation strength or number of hidden layers. The function you're maximising is the model's performance score (such as accuracy or F1), but since the relationship between inputs and output isn't known, it's treated as a black-box function. Because this is a commonly used model, you might benefit from researching best practices or literature to guide your initial search space. Your goal is to find the combination of hyperparameters that yields the highest possible performance.
8	You're optimising an eight-dimensional black-box function, where each of the eight input parameters affects the output, but the internal mechanics are unknown. Your objective is to find the parameter combination that maximises the function's output, such as performance, efficiency or validation accuracy. Because the function is high-dimensional and likely complex, global optimisation is hard, so identifying strong local maxima is often a practical strategy. For example, imagine you're tuning an ML model with eight hyperparameters: learning rate, batch size, number of layers, dropout rate, regularisation strength, activation function (numerically encoded), optimiser type (encoded) and initial weight range. Each input set returns a single validation accuracy score between 0 and 1. Your goal is to maximise this score.

Results

Function	Description
1	Maxima in initial data : 7.71087511e-16 Maxima in my data : 0.0127796427 Net improvement : 0.012779642699999228 Relative improvement : 1,657,353,074,383,080%
2	Maxima in initial data : 0.61120522 Maxima in my data : 0.61521354 Net improvement : 0.00400832 Relative improvement : 0.655805917%
3	Maxima in initial data : -0.03483531 Maxima in my data : -0.0080312 Net improvement : 0.02680411 Relative improvement : 76.94523172%
4	Maxima in initial data : -4.02554228 Maxima in my data : 0.6849177 Net improvement : 4.71045998 Relative improvement : 117.0142965%
5	Maxima in initial data : 1088.85962 Maxima in my data : 8662.4825 Net improvement : 7573.62288 Relative improvement : 695.5554913%
6	Maxima in initial data : -0.71426495 Maxima in my data : -0.20908465 Net improvement : 0.5051803 Relative improvement : 70.727298%
7	Maxima in initial data : 1.3649683 Maxima in my data : 1.7516398 Net improvement : 0.3866715 Relative improvement : 28.3282403%
8	Maxima in initial data : 9.598482 Maxima in my data : 9.89987273 Net improvement : 0.30139073 Relative improvement : 3.13998328%

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